

6. A student says that a simple pendulum oscillates with simple harmonic motion with a period, T , of:

$$T = 2\pi\sqrt{\frac{l}{g}}$$

where l is the length of the pendulum and g is the acceleration due to gravity.

(a) Describe what is meant by a simple pendulum. [1]

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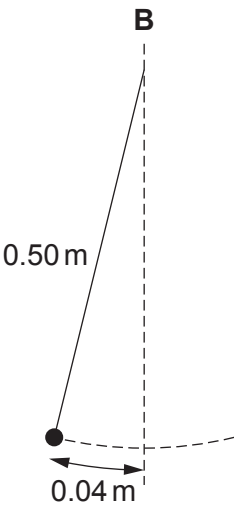
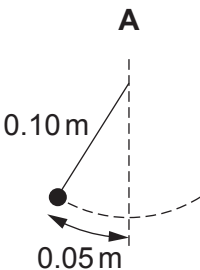
(b) State what is meant by simple harmonic motion. [2]

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(c) The student sets up two experimental systems to investigate simple harmonic motion. System A is a pendulum of length 0.10 m oscillating with an amplitude of 0.05 m. System B is a pendulum of length 0.50 m oscillating with an amplitude of 0.04 m. Give two reasons why system B is better than system A to investigate simple harmonic motion. [2]



Diagrams not to scale.

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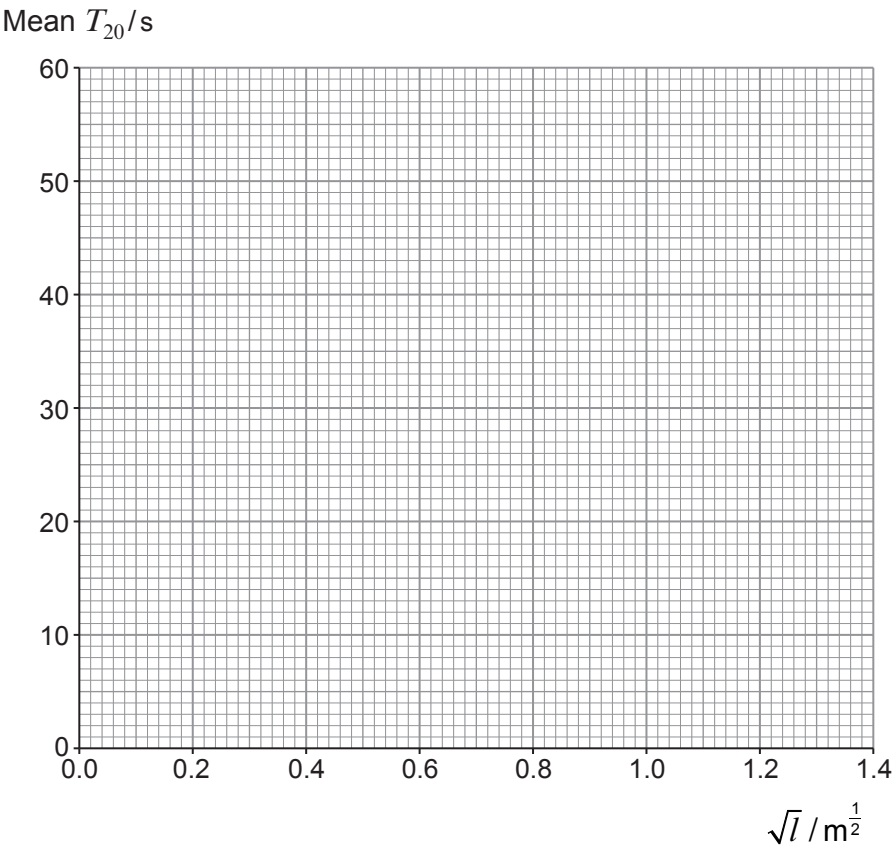
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- (d) Measurements for an oscillating simple pendulum are given in the table below, where l is the length of the pendulum and T_{20} is the time taken for 20 oscillations. For each value of l four values of T_{20} are measured.

l / m	$\sqrt{l} / \text{m}^{\frac{1}{2}}$	T_{20} / s	Mean T_{20} / s	Uncertainty T_{20} / s
0.250	0.50	19, 21, 22, 20	21	2
0.500	0.71	26, 29, 26, 29	28	2
0.750		35, 34, 33, 36		
1.000		42, 39, 40, 41		
1.250		45, 42, 44, 45		
1.500		48, 50, 49, 52		

- (i) **Complete the table** by determining $\sqrt{l}$, the mean time for 20 oscillations, T_{20} and the uncertainty in T_{20} . Values have been inserted for the first two lengths. [3]
- (ii) Plot mean T_{20} against $\sqrt{l}$ on the grid below. Include error bars on the T_{20} axis only. Draw a line of maximum gradient and a line of minimum gradient. [4]



Examiner
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(iii) Justify why error bars for $\sqrt{t}$ are not plotted on the graph. [2]

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(iv) Use the maximum and minimum gradients to determine a mean value for the acceleration due to gravity, g , and its **percentage** uncertainty. [5]

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(e) Another student says that the gradient of the graph would be approximately twice as large if the experiment was carried out on the Moon, where the acceleration is approximately $0.2g$. Investigate if this statement is correct, justifying your answer. [3]

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